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ReMem: A Dynamic Memory Evolution Detector for Zero-Shot Anomaly Detection

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ttps://github.com/fork1917/ReMem

Abstract

Zero-shot anomaly detection (ZSAD) has received increasing attention in industrial manufacturing. Recent approaches mitigate the interference of anomalous samples by constructing purified memory banks. However, conventional one-time static construction heavily depends on unreliable initial evaluations, making the memory bank vulnerable to contamination. To address this limitation, we propose ReMem, a dynamically self-evolving memory bank-driven detector for zero-shot anomaly detection. ReMem first builds an initial memory bank by integrating a Mutual-Scoring Selection Module (MSSM) for score-based filtering and a Dual-Path Feature Enhancement (DPFE) module for robust feature enrichment. An iterative memory evolution mechanism then dynamically refines and expands the memory through repeated MSSM re-evaluation and patch-level selection, effectively suppressing noise and avoiding redundancy. Without any additional training or textual prompts, ReMem achieves 98.7% image-level AUROC and 70.0% pixel-level AP on MVTec AD, and 94.3% image-level AUROC and 48.2% pixel-level AP on VisA, establishing new state-of-the-art performance across both benchmarks.

1. Introduction

Industrial anomaly detection (IAD) aims to identify anomalous regions of products during industrial production (e.g., missing components, surface scratches, stains, and deformations) and is essential for ensuring product quality and production stability. Recently, unsupervised anomaly detection methods have been studied due to the scarcity of anomalous examples [11, 19, 22]. However, these methods typically assume access to large amounts of high-quality normal data, which is often impractical in real-world industrial settings due to data privacy constraints, acquisition costs, and environmental complexity.

In this context, zero-shot anomaly detection (ZSAD), which detects unseen anomalies without task-specific training data, is emerging as a promising direction. Existing ZSAD approaches can be broadly divided into two categories. The first leverages vision-language models (VLMs) and text prompts to describe normal and anomalous patterns [3, 16, 41], while the second constructs a memory bank of normal samples as references for test instances [20]. Compared to the former, the latter is training-free and does

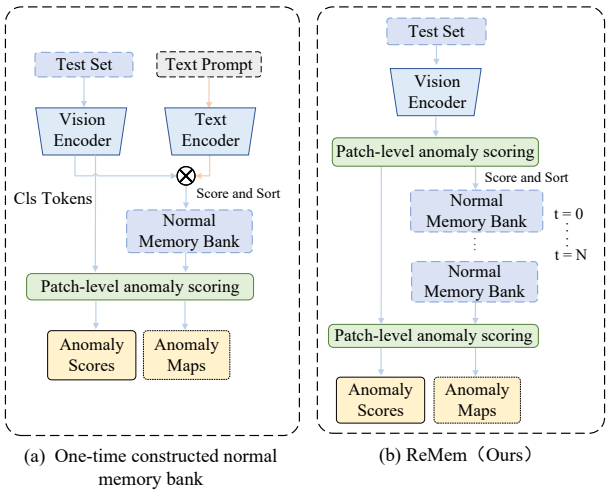


Figure 1. (a) A normal memory bank constructed via a one-time decision is vulnerable to contamination from initial noise. (b) ReMem uses a dynamic, iterative framework (from $t = 0$ to $t = N$) to progressively evolve and purify the memory bank.

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not rely on external textual guidance, but heavily depends on the quality of the memory bank [15]. To the best of our knowledge, existing memory banks are typically built in a one-shot, static manner, making them vulnerable to contamination by anomalous noise and leading to the misidentification of noise-like anomalies.

To address this challenge, we propose ReMem, a zero-shot anomaly detection framework equipped with a dynamically self-evolving memory bank, as illustrated in Fig. 1. Unlike prior work, the proposed framework adopts a dynamic and iterative evolution strategy that progressively refines the memory bank to guide subsequent rounds of sample selection. This design decomposes a single high-stakes decision into a multi-step, well-controlled process, effectively mitigating biases introduced by early one-shot construction and expanding the memory bank's coverage in a more reliable manner.

Specifically, we employ a pre-trained visual encoder to extract feature representations, which are dynamically stored and refined within the memory bank. In the initial stage, a Mutual-Scoring Selection Module (MSSM) is introduced to score and select features that are more likely to belong to normal samples based on inter-sample similarity. The selected features are then processed by a Dual-Path Feature Enhancement (DPFE) module, which decouples texture and object representations through multi-scale multi-layer pooling and a PCA-based mask, respectively. During the memory bank update stage, the MSSM module re-evaluates candidate normal features using the continuously updated DPFE-enhanced memory bank. To reduce redundancy, only a small subset of high-confidence features is added at each iteration. Finally, during inference, the DPFE module compares the test patch embeddings with the evolved memory bank to accurately localize anomalous regions. The contributions of this paper are summarized as follows:

- We propose **ReMem**, a zero-shot anomaly detection framework with a dynamic self-evolving memory bank.
- ReMem introduces a dynamic self-evolving memory bank to iteratively and reliably expand the bank, couples it with a patch-level selection strategy to avoid redundancy, and employs dual-path feature enhancement (DPFE) to adaptively suppress noise and accentuate fine details.
- Our approach achieves state-of-the-art performance on MVTec AD and VisA: 98.7% image-level AUROC and 95.5% pixel-level PRO on MVTec AD; 94.3% image-level AUROC and 94.7% pixel-level PRO on VisA.

2. Related Work

2.1. Industrial Anomaly Detection

IAD can be broadly divided into three groups: reconstruction-based, embedding-based, and synthesis-based approaches.

Reconstruction-based methods [4, 12, 14, 39] rely on the assumption that a model can accurately reconstruct normal samples but fails to reconstruct anomalous ones. With the rapid advancement of generative models, variants built upon VAEs [21, 24], GANs [1, 32], and diffusion models [34, 40] have become widely adopted.

Embedding-based methods extract features from pre-trained networks to map normal data into a compact feature space, often combined with techniques such as knowledge distillation [31] or memory-bank retrieval [28]. Anomalies are then detected by measuring the distance between test and normal features. For instance, SPADE [8] introduced a memory bank of normal features and employed k-nearest neighbor (k-NN) distances for anomaly detection and localization, while PatchCore [29] further applied greedy coreset subsampling to reduce redundancy and balance accuracy with runtime efficiency. Both types are trained solely on normal data and, therefore, fail to model the anomaly distribution.

Synthesis-based methods [13, 37] address this limitation by generating synthetic anomalies through data augmentation [2, 17, 35, 36, 38] or by injecting feature-space perturbations [5, 22, 27], encouraging the model to learn more discriminative decision boundaries.

2.2. Zero-Shot Anomaly Detection

ZSAD aims to detect unseen anomalies without task-specific training data.

One branch leverages CLIP for image-text alignment. As an early pioneer, WinCLIP [16] computes embedding similarities between image patches and hand-crafted text prompts describing normal and anomalous states. APRIL-GAN [6] and CLIP-AD [7] further enhance CLIP for ZSAD by tuning additional projection layers with labeled auxiliary anomaly data. Building on this idea, AnomalyCLIP [41] introduces learnable text prompts to adapt vision-language models (VLMs) to ZSAD, inspiring a wave of prompt-centric approaches. However, these methods heavily rely on external text prompts and extensive prompt engineering, and CLIP's text embeddings often struggle to separate normal from anomalous semantics [18, 23, 33].

Another line of work detects anomalies through mutual scoring among unlabeled test images. This paradigm originated with MuSc [20], which observed that normal patches tend to have many similar counterparts in other test images, whereas anomalous patches have few. MuSc introduces a mutual scoring mechanism in which test images assign anomaly scores to one another, achieving strong results. Inspired by this, AnomalyDINO [9], originally a few-shot method, proposed a simplified zero-shot batched extension using the same mechanism. Subsequently, PA-CLIP [26] addresses pseudo anomalies with a dual memory bank de-

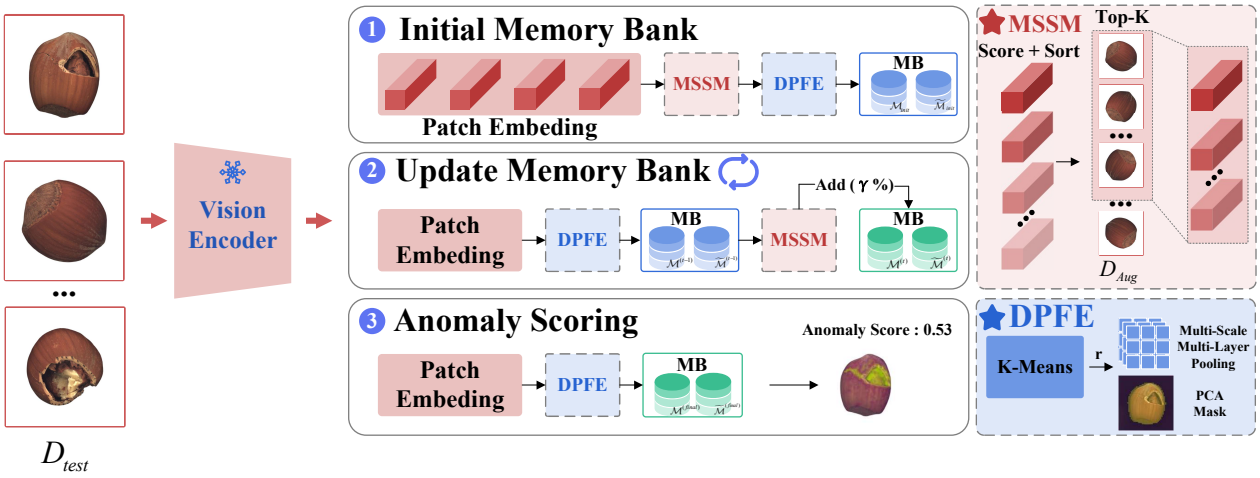


Figure 2. Overview of the ReMem framework. It comprises three components: Initial Memory Bank, Update Memory Bank, and Anomaly Scoring.

sign—comprising a normal bank and a full bank—to reduce background noise and false positives. PA-CLIP uses a fixed text prompt to pre-screen samples for the normal bank, partially mitigating noise issues in MuSc. However, this pre-screening relies on a one-time static decision from CLIP's coarse-grained text branch, which may still admit anomalous noise, contaminating the memory bank and degrading downstream detection.

It is evident that, whether based on mutual scoring or memory-bank screening, existing methods are fundamentally constrained by a static paradigm. In this paper, we propose a dynamic memory-bank self-evolving ZSAD framework, replacing the one-time static decision with an iterative process, where the current memory bank serves as the reference for re-evaluating candidate samples and is progressively refined at each round.

3. Method

3.1. Overview

As shown in Fig. 2, the ReMem detection process consists of three stages: initial memory bank, update memory bank, and anomaly scoring.

3.2. Initial Memory Bank

To begin, we adopt a pre-trained DINOv2 [25] as the vision encoder to extract features. The MSSM is then used to identify high-confidence candidate-normal samples from the unlabeled test set, while the DPFE module routes their features through object and texture paths to construct the corresponding initial memory banks, $\mathcal{M}_{\text{init}}$ and $\tilde{\mathcal{M}}_{\text{init}}$. For clarity, we next provide a detailed description of the MSSM and DPFE modules.

Mutual-Scoring Selection Module. Given the test set D_{test}, we first construct a temporary memory bank $\mathcal{M}_i$ for each sample I_i, consisting of patch features from all other samples in the test set:

$$\mathcal{M}_i = \mathcal{M}_{test} \setminus P^{(I_i)}, \tag{1}$$

where $\mathcal{M}_{test}$ is the patch feature set of D_{test} and $P^{(I_i)}$ is the patch feature set of I_i. With $\mathcal{M}_i$ as the reference memory bank, the anomaly score of each patch feature of I_i can be computed as:

$$s^{(I_i)} = \frac{1}{\mathcal{N}(\mathcal{M}_i(\epsilon))} \sum_{\mathcal{N}(\mathcal{M}_i(\epsilon))} d\left(p^{(I_i)}, \mathcal{M}_i(\epsilon)\right), \tag{2}$$

where, $d(\cdot,\cdot)$ is the cosine distance and $\mathcal{N}(.)$ is the patch number of $(.)$. $\mathcal{M}_i(\epsilon) \subset \mathcal{M}_i$ denotes the top 0.1% nearest reference patches used to compute the patch-level score. We then aggregate the patch-level scores into an image-level anomaly score for sample ranking:

$$S^{(I_i)} = \frac{1}{\mathcal{N}(P^{(I_i)})} \sum_{\mathcal{N}(P^{(I_i)})} s^{(I_i)}(\xi), \tag{3}$$

where $s^{(I_i)}(\xi)$ denotes the top 1% highest patch-level anomaly scores of image I_i used for image-level aggregation. Finally, we select the top-K samples with the lowest image-level anomaly scores as candidate-normal samples and use their features to construct the initial memory bank.

Dual-Path Feature Enhancement. In IAD, object-class detection is often hindered by background noise [38],

whereas texture-class detection requires capturing fine-grained local details [29]. To address these two cases in a unified manner, we design DPFE as a dual-path routing module. It first obtains a coarse foreground proposal by applying K-Means clustering with $C = 2$ to patch features and then refines it with simple morphological operations before routing the sample to either the object path or the texture path.

Let r denote the foreground mask area ratio, which is used as the routing cue in DPFE. If $\mathrm{r} < \tau_r$, the sample is routed to the object path, where a PCA-based masking strategy is applied and only foreground patch features are retained to construct the initial object memory bank $\mathcal{M}_{\text{init}}$.

Conversely, when $\mathrm{r} \geq \tau_r$, the sample is routed to the texture path, where Multi-Layer Multi-Scale Pooling (MLMP) is used to enhance fine-grained local details. Specifically, following the formulation in the supplementary material, we aggregate features from multiple shallow encoder layers L and pooling scales $\mathcal{R}$ to obtain the enhanced texture features $\tilde{\mathbf{P}}^{(I)}$, which are used to construct the texture memory bank $\tilde{\mathcal{M}}_{\text{init}}$. In summary, MSSM identifies candidate-normal samples from the unlabeled test set, and DPFE determines how their features are processed before being added to the initial memory bank.

3.3. Update Memory Bank

Let $\mathcal{D}^{(t)}_{\text{cand}}$ denote the candidate sample set at iteration t, i.e., the samples not yet included in the current memory bank. Using the current memory bank as the reference, we employ MSSM to re-score all samples in $\mathcal{D}^{(t)}_{\text{cand}}$ and select a subset of candidate-normal samples, denoted by $\mathcal{D}^{(t)}_{\text{normal}}$. For object-path samples, the PCA-based masking strategy in DPFE is further applied to retain object-related patch features before the memory bank update.

As highlighted in PatchCore [29], incorporating all patch features from every selected sample introduces substantial redundancy. To reduce this redundancy, we use anomaly scores as a diversity-oriented criterion for patch selection, motivated by the intuition of k-NN density estimation. In this view, the local density of a patch feature can be written as

$$\hat{p} \propto \frac{1}{(r_1)^d}, \tag{4}$$

where r_1 is the nearest-neighbor distance between this patch feature and those in the current memory bank, and d is the feature dimension. Combined with Eq. 2, this density view provides an intuitive interpretation of the anomaly score:

$$s \propto \left(\hat{p}\right)^{-1/d}. \tag{5}$$

Accordingly, within the MSSM-filtered candidate-normal samples in $\mathcal{D}^{(t)}_{\text{normal}}$, a higher anomaly score indicates a less redundant yet still informative patch relative to the current memory bank. Based on this intuition, we retain the top $\gamma\%$ highest-scoring patches from each sample in $\mathcal{D}^{(t)}_{\text{normal}}$ and use them to form the update subsets $\Delta\mathcal{M}^{(t)}$ and $\Delta\tilde{\mathcal{M}}^{(t)}$ for the object and texture paths, respectively. The memory banks are then updated as:

$$\mathcal{M}^{(t)} = \mathcal{M}^{(t-1)} \cup \Delta\mathcal{M}^{(t)}, \tag{6}$$

where $\Delta\mathcal{M}^{(t)}$ denotes the selected patch features extracted from the object-path samples in $\mathcal{D}^{(t)}_{\text{normal}}$.

The texture memory bank is updated in the same manner:

$$\tilde{\mathcal{M}}^{(t)} = \tilde{\mathcal{M}}^{(t-1)} \cup \Delta\tilde{\mathcal{M}}^{(t)}, \tag{7}$$

where $\Delta\tilde{\mathcal{M}}^{(t)}$ denotes the selected enhanced patch features extracted from the texture-path samples in $\mathcal{D}^{(t)}_{\text{normal}}$.

We stop the iterative update when the accumulated number of added samples reaches $\tau\%$ of the test set size, so that τ controls the final memory expansion budget. The resulting memory banks are denoted by $\mathcal{M}^{(\text{final})}$ and $\tilde{\mathcal{M}}^{(\text{final})}$ for the object and texture paths, respectively.

3.4. Anomaly Scoring

In the final stage, we perform anomaly scoring and localization for a test sample I using the evolved memory banks $\mathcal{M}^{(\text{final})}$ and $\tilde{\mathcal{M}}^{(\text{final})}$.

For object-class, we first apply the PCA-based masking strategy in DPFE to extract foreground patch features. The anomaly score of each foreground patch is then computed with respect to $\mathcal{M}^{(\text{final})}$ using Eq. 2. The patch-level scores are subsequently aggregated via Eq. 3 to obtain the image-level anomaly score. For pixel-level localization, the patch-score map is bilinearly upsampled to the original image resolution and further refined using Gaussian smoothing [10].

For texture-path samples, the scoring pipeline is slightly different. Image-level anomaly scoring uses the base evolved memory bank $\mathcal{M}^{(\text{final})}$ to preserve global consistency, whereas pixel-level localization uses the enhanced memory bank $\tilde{\mathcal{M}}^{(\text{final})}$ together with the DPFE-enhanced features $\tilde{\mathbf{P}}^{(I)}$ to better capture local texture details. Patch anomaly maps are computed using Eq. 2 and undergo the same upsampling and Gaussian smoothing post-processing.

4. Experiments

4.1. Implementation Details

Datasets. Our experiments are conducted on the MVTec AD and VisA datasets. MVTec AD contains 10 object and 5 texture categories, with a total of 1,725 high-resolution RGB images in the test set. VisA consists of 12 object categories, with 962 normal and 1,200 anomalous images in

its test split. The anomalies in VisA exhibit high diversity, covering both surface and structural defects.

Baselines and Metrics. For the anomaly classification task, we report 3 image-level metrics: Area Under the ROC Curve (AUROC), F1 score at the optimal threshold (F1-max), and Average Precision (AP). For anomaly segmentation, we report pixel-level AUROC, AP, F1-max, and Per-Region Overlap (PRO). The compared baseline methods in this paper are as follows: PatchCore[29], WinCLIP [16], APRIL-GAN [6], AnomalyDINO[9], AdaCLIP[3], AnomalyCLIP[41], Crane[30], MuSc[20], PA-CLIP[26].

Experiment Settings. We adopt ViT-S/14 from the pretrained DINOv2 model as the visual encoder. During preprocessing, each input image is resized such that its shorter edge is scaled to 672 pixels while preserving the aspect ratio. For the MSSM, we set Top-$K = 3$ and the patch selection ratio $\gamma = 10\%$. For the DPFE module, we extract features from encoder layers $L = 6, 9$ with pooling scales $\mathcal{R} = 1, 5$, and set the routing threshold to $\tau_r = 0.5$. The final anomaly map is bilinearly upsampled to the original resolution. All experiments are performed on a single NVIDIA A6000 GPU.

4.2. Main Results

Quantitative Results. Tables 1 and 2 present the quantitative results of various zero-shot and few-shot methods on the MVTec AD and VisA datasets. Our proposed ReMem achieves state-of-the-art performance across both image-level and pixel-level metrics. On MVTec AD, ReMem surpasses the second-best ZSAD method by 0.9% AUROC, 5.0% F1, 6.6% AP, and 1.8% PRO at the pixel level. Compared with the few-shot (4-shot) baseline AnomalyDINO, ReMem shows additional gains of 1.0% in image-level AUROC and 10.0% in pixel-level AP.

On the VisA dataset, ReMem improves over the second-best ZSAD method by 0.9% in image-level AUROC and 2.2% in pixel-level AP. It also outperforms the 4-shot AnomalyDINO by 1.7% in image-level AUROC and 3.9% in pixel-level AP. Notably, MuSc exhibits low AP scores due to severe contamination by anomalous noise, which induces many false positives. In contrast, ReMem's dynamic self-evolving memory bank effectively suppresses such noise, resulting in significantly fewer false alarms.

Fig. 3 provides a visual comparison between ReMem and two representative baselines, AnomalyCLIP and MuSc, on MVTec AD and VisA. As shown, ReMem produces more precise anomaly-localization heatmaps. These qualitative results demonstrate that the proposed dynamic memory-bank evolution mechanism constructs a cleaner and more faithful representation of normal patterns, enabling better separation between true anomalies and complex normal structures.

4.3. Ablation Study and Analysis

Ablation of different components. We analyze the contribution of each component in ReMem on the MVTec AD and VisA datasets. As shown in Table 3, the baseline corresponds to the batched zero-shot variant of AnomalyDINO used within MSSM. Adding the initial memory bank with purified normal representations yields substantial improvements of +3.4% in image-level AUROC and +9.6% in pixel-level PRO on MVTec AD. However, its slightly lower image-level performance on VisA highlights the importance of incorporating the dynamic self-evolving memory bank. Once the dynamic mechanism is enabled, VisA image-level AUROC increases by 2.7%, along with a 1.4% PRO gain, demonstrating that iterative refinement effectively broadens and stabilizes the normal representation space.

Removing the DPFE module results in clear performance degradation, including a 4.1% drop in image-level AUROC on VisA and a 2.1% decrease in pixel-level PRO on MVTec AD, confirming its crucial role in enhancing object and texture representations. The best results on both MVTec AD and VisA are achieved when all components are jointly enabled. Overall, these findings validate the superiority of our framework and the necessity of each component in achieving robust zero-shot anomaly detection.

Discussion of the dynamic self-evolving memory bank. Fig. 4 illustrates the evolution of detection performance as the dynamic self-evolving memory bank updates across iterations on the MVTec AD and VisA datasets. Image-level metrics exhibit a clear upward trend as the memory bank grows, demonstrating that progressively integrating diverse normal features steadily broadens the coverage of normal patterns, thereby improving anomaly detection performance.

The star markers highlight the results of a one-shot, statically constructed memory bank, which performs notably worse across all metrics (AUROC, F1, AP) on both datasets. This contrast underscores the necessity and advantage of dynamic evolution: rather than relying on a single high-stakes decision, our iterative strategy decomposes memory bank construction into multiple controlled steps, mitigating early-stage errors and yielding a more reliable and comprehensive representation of normality.

Analysis of patch-level selection strategy. As shown in Table 4, we evaluate the impact of different thresholds $\gamma\%$ in the patch-level selection strategy. Overall, ReMem achieves the best overall performance at $\gamma = 10\%$. Building on this setting, we further analyze the effect of different patch-selection intervals. Although patches from earlier intervals exhibit higher anomaly scores—making them closer to anomalous patterns—they also contain the most informative features for constructing a diverse memory bank.

The results in Fig. 5, where each point corresponds to the starting position of a 10%-wide selection interval, reveal

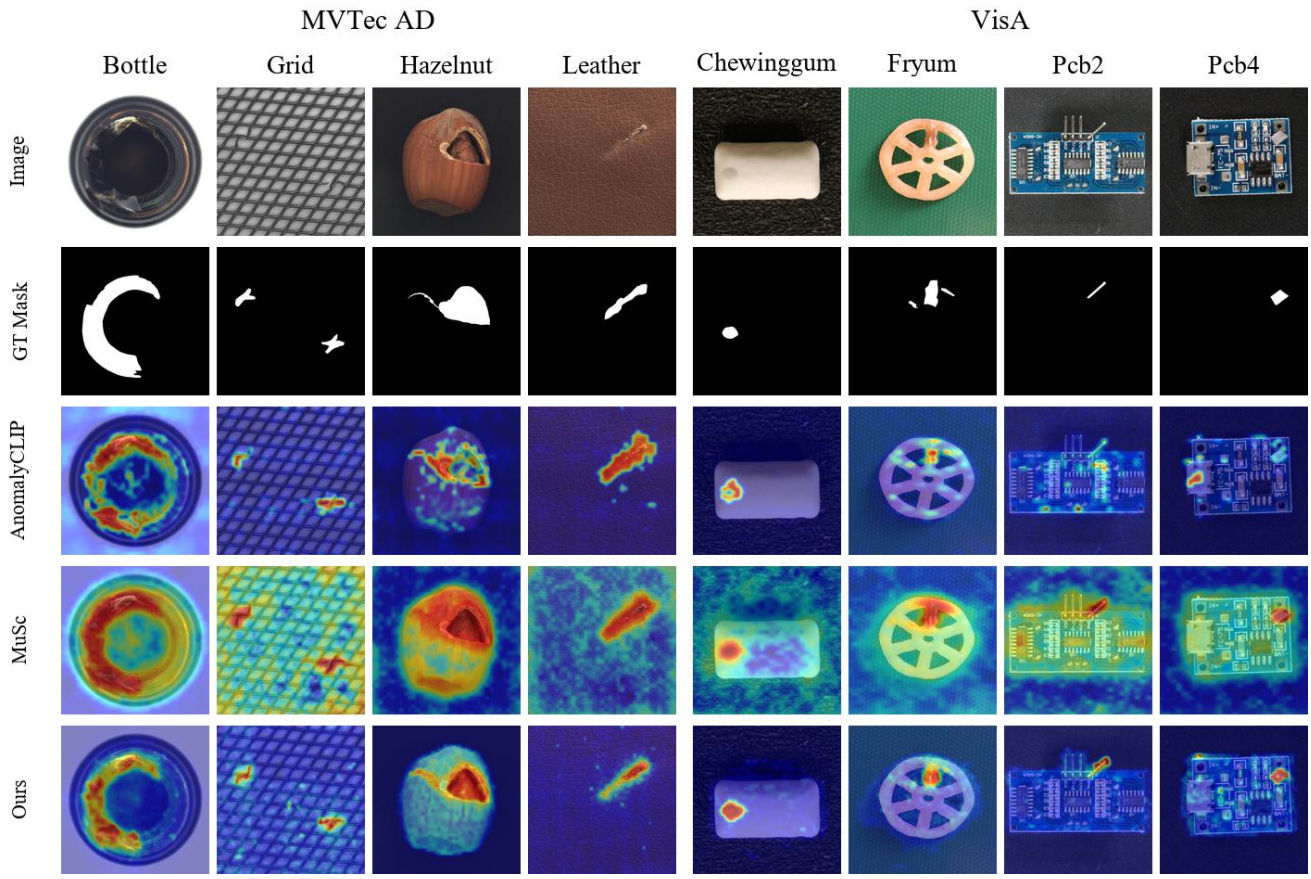


Figure 3. Visualization of anomaly detection results on the MVTec AD and VisA datasets. Each column corresponds to a different category: the first row shows the input image, the second row shows the ground-truth mask, and the third to fifth rows show the detection results of AnomalyCLIP, MuSc, and ReMem, respectively.

Table 1. Quantitative results on MVTec AD (%). Best results are **boldfaced**, second-best results are underlined, and our row is shaded.

Methods	Setting	AUROC-cls	F1-max-cls	AP-cls	AUROC-segm	F1-max-segm	AP-segm	PRO-segm
PatchCore[29]	4-shot	88.8	92.6	94.5	94.3	55.0	–	84.3
WinCLIP[16]	4-shot	95.2	94.7	97.3	96.2	51.7	–	88.0
APRIL-GAN[9]	4-shot	92.8	92.8	96.3	95.9	56.9	–	91.8
AnomalyDINO[9]	4-shot	97.7	96.6	98.7	97.2	61.8	60.0	93.4
WinCLIP[30]	0-shot	91.8	92.9	96.5	85.1	31.7	–	64.6
AdaCLIP[3]	0-shot	89.2	90.6	95.7	88.7	43.4	–	37.8
AnomalyCLIP[41]	0-shot	91.5	92.7	96.2	91.1	39.1	–	81.4
Crane[30]	0-shot	93.9	93.6	97.6	91.2	43.8	–	88.1
MuSc[20]	0-shot	97.8	97.5	99.1	97.3	62.6	62.7	93.8
PA-CLIP[26]	0-shot	98.4	97.8	99.3	97.5	63.0	63.4	93.7
ReMem(Ours)	0-shot	**98.7**	**98.1**	**99.5**	**98.4**	**68.0**	**70.0**	**95.5**

a clear trend across both MVTec AD and VisA. The best image-level AUROC is obtained when the interval starts at 0, i.e., selecting the top 10% highest-scoring patches. As the interval shifts toward lower-scoring patches, performance steadily declines, with a more pronounced drop on VisA. This observation strongly supports that, within our iterative evolution framework, the advantages of incorporating high-scoring patches far outweigh the potential risk of including features that may resemble anomalies.

Ablation of the dual paths in DPFE. As shown in Ta-

Table 2. Quantitative results on VisA (%). Best results are **boldfaced**, second-best results are underlined, and our row is shaded.

Methods	Setting	AUROC-cls	F1-max-cls	AP-cls	AUROC-segm	F1-max-segm	AP-segm	PRO-segm
PatchCore[29]	4-shot	85.4	84.3	87.5	96.8	43.9	–	89.4
WinCLIP[16]	4-shot	87.3	84.2	88.8	97.2	47.0	–	87.6
APRIL-GAN[9]	4-shot	92.6	88.4	**94.5**	96.2	40.0	–	90.2
AnomalyDINO[9]	4-shot	92.6	88.8	92.9	98.2	49.4	44.3	94.1
WinCLIP[30]	0-shot	79.6	80.7	81.1	79.6	14.8	–	56.8
AnomalyCLIP[41]	0-shot	82.1	80.4	85.4	95.5	28.3	–	87.0
Crane[30]	0-shot	83.6	81.2	86.7	95.3	30.2	–	90.6
AdaCLIP[3]	0-shot	85.8	83.1	79.0	95.5	37.7	–	72.9
MuSc[20]	0-shot	92.8	89.5	93.5	**98.8**	48.8	45.1	92.7
PA-CLIP[26]	0-shot	93.4	89.7	93.9	**98.8**	50.3	46.0	92.4
ReMem(Ours)	0-shot	**94.3**	**90.8**	**94.5**	98.2	**52.2**	**48.2**	**94.7**

Table 3. Ablation results on MVTec AD and VisA. [A] baseline; [B] initial memory bank without DPFE; [C] dynamic self-evolving memory bank without DPFE; [D] initial memory bank with DPFE; [E] full method. The full-method row is shaded and best results are **boldfaced**.

Methods	**MVTec AD**		**VisA**	
	Image-level (AUROC, AP)	Pixel-level (AUROC, PRO)	Image-level (AUROC, AP)	Pixel-level (AUROC, PRO)
[A]	(94.2, 97.3)	(95.6, 83.8)	(90.6, 92.0)	(97.6, 89.9)
[B]	(97.6, 98.8)	(97.2, 93.4)	(87.5, 87.2)	(98.0, 92.0)
[C]	(98.1, 99.1)	(97.2, 93.4)	(90.2, 90.3)	(98.0, 93.4)
[D]	(98.3, 99.2)	(98.1, 95.4)	(91.3, 92.3)	(98.1, 93.9)
[E]	(**98.7**, **99.5**)	(**98.4**, **95.5**)	(**94.3**, **94.5**)	(**98.2**, **94.7**)

Table 4. Impact of the patch-selection threshold (γ) on VisA. The best-performing setting is shaded and best results are firstboldfaced.

γ%	**Image-level** (AUROC, AP)	**Pixel-level** (AUROC, PRO)
5%	(93.8, 94.2)	(98.0, 94.3)
10%	(**94.3**, **94.5**)	(**98.2**, 94.7)
30%	(94.1, 94.4)	(98.1, 94.8)
50%	(94.1, 94.3)	(98.1, **95.0**)
100%	(94.2, 94.5)	(98.2, 95.0)

Table 5. Ablation of the PCA-based masking and MLMP modules on MVTec AD. The baseline corresponds to the full method without DPFE. The complete module row is shaded and best results are firstboldfaced.

Modules	**Image-level** (AUROC, AP)	**Pixel-level** (AUROC, PRO)
Baseline	(98.1, 99.1)	(97.2, 93.4)
PCA MASK	(98.7, 99.5)	(97.6, 93.8)
MLMP	(98.1, 99.1)	(98.0, 95.2)
PCA MASK + MLMP	(**98.7**, **99.5**)	(**98.4**, **95.5**)

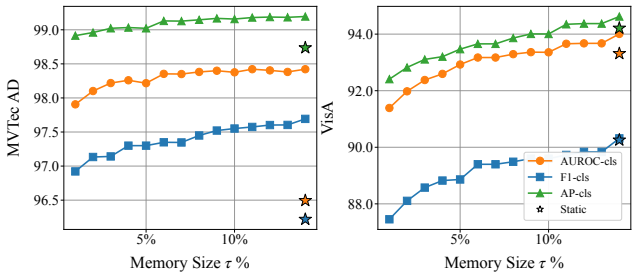


Figure 4. Performance variation of image-level metrics (AUROC, F1-max, AP) with respect to the memory bank expansion ratio τ% on MVTec AD (left) and VisA (right). The star marker indicates the performance obtained using the one-shot static memory bank construction.

ble 5, we conduct an ablation study on the dual-path design of DPFE. The results show that incorporating either the PCA-based masking or the MLMP pathway individually outperforms the baseline, confirming the usefulness of both components. Specifically, the PCA MASK pathway improves image-level AUROC by 0.6% and AP by 0.4% by suppressing background regions. The MLMP pathway, designed to enhance texture features, also yields clear gains, increasing pixel-level AUROC from 97.2% to 98.4% and PRO from 93.4% to 95.5%.

The best performance is achieved when both pathways are combined. The complete DPFE module, integrating PCA MASK and MLMP, delivers the highest overall results. These findings validate the effectiveness and necessity of the dual-path design.

5. Conclusion

This paper addresses the limitations of static memory banks in zero-shot anomaly detection by proposing ReMem, a

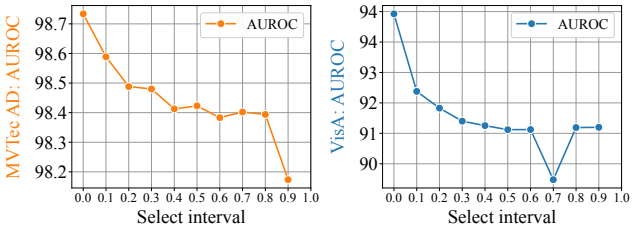


Figure 5. Image-level AUROC under different patch-selection intervals on the MVTec AD (left) and VisA (right) datasets. Each point corresponds to the starting value of a selection interval, where all intervals share a fixed length of 10%.

novel framework with a dynamic self-evolving memory bank. Unlike conventional zero-shot methods that rely on potentially unreliable initial memory banks, ReMem employs a multi-stage construction process. It begins with a purified initial memory bank built via the MSSM for robust sample filtering, complemented by the DPFE module to separate and refine feature representations. Through iterative memory evolution, involving repeated re-evaluation and anomaly-score–driven patch selection, ReMem progressively expands memory diversity while suppressing noise and reducing redundancy. Without additional training or text prompts, ReMem sets a new state-of-the-art, surpassing existing zero-shot methods and most few-shot approaches.

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